APPELLO 12 FEBBRAIO 2020

ESERCIZIO 1:

a) SIA $\vec{E}$ CAMPO ELETTRICO UNIFORME

$\Rightarrow \vec{F_1} = q\vec{E} = -\vec{F_2}$ LA RISULTANTE

È DUNQUE $\vec{R} = 0$.

$\vec{F_1}$ ED $\vec{F_2}$ FORMANO PERÒ UNA COPPIA

$\vec{\tau} = \vec{a} \times q\vec{E} = q\vec{a} \times \vec{E} = \vec{p} \times \vec{E}$

CON $\vec{p} = q\vec{a}$ MOM. DI DIPOLO.

SIA ORA $\vec{E}$ CAMPO ELETTRICO NON UNIFORME

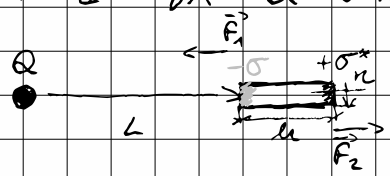
$\vec{F_1} = -q\vec{E}$, $\vec{F_2} = q(\vec{E} + \Delta\vec{E}) \Rightarrow$

$\vec{R} = -q\vec{E} + q\vec{E} + q\Delta\vec{E} = q\Delta\vec{E}$

$\nabla(\vec{E} \cdot \vec{a}) = \Delta\vec{E} \Rightarrow \vec{R} = \nabla(\vec{p} \cdot \vec{E})$

CILINDRETTO DIELETTRICO $h \gg R$

DISTANZA L DA Q CARICA PUNTIFORME. $\vec{\tau} = 0$.



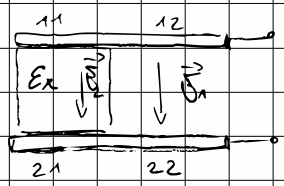
$F_1 > F_2$

b) ESSENDO $\tau = 0$ IL CILINDRETTO HA IL SUO ASSE DIRETTO RADIALMENTE RISPETTO LA CARICA PUNTIFORME, LA RISULTANTE AGISCE RADIALMENTE PUNTANDO VERSO LA CARICA.

c) L'UNICO EQUILIBRIO È EQUILIBRIO INSTABILE RAGGIUNTO QUANDO IL CILINDRETTO TOCCHERÀ LA CARICA Q.

ESERCIZIO 2:

C PIANO, S; d V_0 Q_0 GENERATORE STACC.
INSERITA LASTRA $S/2$, d, ε_r



$Q_0 = C V_0$

Q_0 SI CONSERVA DOPO L'INSERIMENTO, C VARIA

$$C_0 = \frac{\varepsilon_0 S}{d} \qquad C_{eq} = \frac{\varepsilon_0 S}{2d} + \frac{\varepsilon_0 \varepsilon_r S}{2d}$$

a) $$V_2 = \frac{\frac{\varepsilon_0 S V_0}{d}}{\frac{\varepsilon_0 S}{2d} + \frac{\varepsilon_0 \varepsilon_r S}{2d}} = \frac{V_0}{\frac{1}{2} + \frac{\varepsilon_r}{2}} = \frac{2V_0}{1+\varepsilon_r}$$

b) $$U_i = \frac{1}{2} Q_0 V_0 = \frac{1}{2} \frac{\varepsilon_0 S}{d} V_0^2$$

$$U_f = \frac{1}{2} Q_0 V_2 = \frac{1}{2} \frac{\varepsilon_0 S V_0}{d} \cdot \frac{2V_0}{1+\varepsilon_r} = \frac{\varepsilon_0 S}{(1+\varepsilon_r) d} V_0^2$$

$$\Delta U = U_f - U_i = \frac{\varepsilon_0 S V_0^2}{d} \left(\frac{1}{1+\varepsilon_r} - \frac{1}{2} \right)$$

$\varepsilon_r > 1 \Rightarrow \Delta U < 0$.

c) $\vec{E}_1 = \vec{E}_2 = \vec{E}$ $\vec{D}_1 = \varepsilon_0 \vec{E}$; $\vec{D}_2 = \varepsilon_0 \varepsilon_r \vec{E}$

$\vec{D} \cdot \vec{n} = \sigma_{lib}$ PER TEO. DI COULOMB.

$$\vec{E} = -\nabla V_2 = \frac{V_2}{d} \vec{u}_x$$

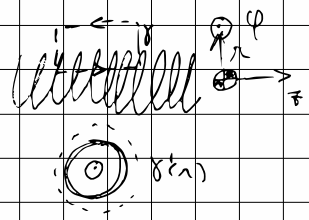
$$\Rightarrow \sigma_{11} = \varepsilon_0 \varepsilon_r \frac{V_2}{d} \; ; \quad \sigma_{12} = \varepsilon_0 \frac{V_2}{d}$$

$$\sigma_{21} = -\varepsilon_0 \varepsilon_r \frac{V_2}{d} \; ; \quad \sigma_{22} = -\varepsilon_0 \frac{V_2}{d}$$

ESERCIZIO 3:

SOLENOIDE INFINITO, a; n $\quad i(t) = i_0 \sin(\omega t)$

a) $\oint_\gamma \vec{B} \cdot d\vec{l} = \mu_0 i c$



$\vec{B} = B(t)\vec{u}_z = \mu_0 n i(t) \vec{u}_z$

$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$\Rightarrow \oint_{\gamma'(r)} \vec{E} \cdot d\vec{l} = -\frac{\partial \Phi_{S(r)}(\vec{B})}{\partial t}$

- $r < a$: $\vec{E} = E(t, r)\vec{u}_\varphi = -\frac{\partial \Phi_{S(r)}(\vec{B})}{\partial t}\vec{u}_\varphi$

$$\Phi_{S(r)}(\vec{B}) = \mu_0 n \int_0^r \int_0^{2\pi} i_0 \sin(\omega t)\, r\, dr\, d\theta =$$

$$= -\frac{\pi i_0}{\omega} \cos(\omega t)\, r^2 \cdot \mu_0 n$$

$$-\frac{\partial \Phi_{S(r)}(\vec{B})}{\partial t} = -\pi i_0 \sin(\omega t)\, r^2 \cdot \mu_0 n$$

$$\vec{E} = -\frac{i_0}{2} \sin(\omega t)\, r\, \vec{u}_\varphi \cdot \mu_0 n$$

- $r > a$: $\vec{E} = E(t, r)\vec{u}_\varphi$

$$\Phi_{S(a)}(\vec{B}) = \int_0^a \int_0^{2\pi} i_0 \sin(\omega t)\, r\, dr\, d\theta\, \mu_0 n =$$

$$= -\frac{\pi i_0}{\omega} \cos(\omega t)\, a^2 \cdot \mu_0 n$$

$$-\frac{\partial \Phi_{S(a)}(\vec{B})}{\partial t} = -\pi i_0 \sin(\omega t)\, a^2 \cdot \mu_0 n$$

$$\vec{E} = -\frac{i_0}{2} \sin(\omega t) \frac{a^2}{r} \vec{u}_\varphi \cdot \mu_0 n$$

b) $a = 5\,cm$, $n = 200$, $i_0 = 10\,A$, $\omega = 100\,\frac{rad}{s}$

$\mu_0 = 4\pi \cdot 10^{-7}\,\frac{N}{A^2}$

$$\xi_{MAX} = \xi(t; a) = \frac{i_0}{2} \sin(\omega t)\, a \cdot \mu_0 n$$

$$\omega t = \frac{\pi}{2} + k\pi \Rightarrow t = \frac{\pi}{200} + \frac{k}{100}\pi$$

$$\xi_{MAX}\left(\frac{\pi}{200} + \frac{k}{100}\pi; a\right) = 4\pi \cdot 10^{-7} \cdot 200 \cdot 0{,}05 \cdot 0{,}5 \cdot 10 =$$
$$= 200\pi \cdot 10^{-7}\,\frac{N}{C}$$

c) PERCHÉ $\vec{\xi}$ È MASSIMO PROPRIO IN $r = a$?

ESERCIZIO 4:

a) δ_{MIN} È TALE PER CUI $\frac{d\delta}{di} = 0$

$$\Rightarrow \frac{d\delta}{di} = 1 + \frac{di'}{di} = 0 \quad \Rightarrow \frac{di'}{di} = 0$$

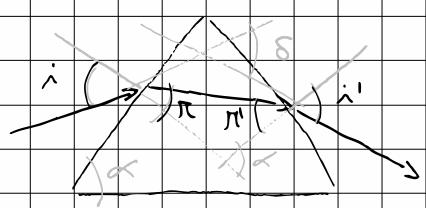
DALLE RELAZIONI DI SNELL:

$$\cos i \, di = n \cos r \, dr \qquad \cos i' \, di' = n \cos r' \, dr'$$

$$r + r' = \alpha \quad \Rightarrow \quad dr = -dr'$$

$$\Rightarrow \frac{di'}{di} = -\frac{\cos i \cos r'}{\cos i' \cos r} = -1$$

a) $\Rightarrow \quad i = i' \quad \wedge \quad r = r'$



ESSENDO $\sin i = n \sin r$ DALLE LEGGI DI SNELL.

$\sin i' = n \sin r'$

$r + r' = \alpha$ E $\delta = (i - r) + (i' - r') =$

$= i + i' - \alpha$

$$\delta_{MIN} = 2i - \alpha$$

c) SPETTROMETRO / MONOCROMATORE A PRISMA